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Microarray-based survey of repetitive genomic sequences in *Vicia* spp.

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Abstract

A modified DNA microarray-based technique was devised for preliminary screening of short fragment genomic DNA libraries from three *Vicia* species (*V. melanops*, *V. narbonensis*, and *V. sativa*) to isolate representative highly abundant DNA sequences that show different distribution patterns among related legume species. The microarrays were sequentially hybridized with labeled genomic DNAs of thirteen *Vicia* and seven other Fabaceae species and scored for hybridization signals of individual clones. The clones were then assigned to one of the following groups characterized by hybridization to: (1) all tested species, (2) most of the *Vicia* and *Pisum* species, (3) only a few *Vicia* species, and (4) preferentially a single *Vicia* species. Several clones from each group, 65 in total, were sequenced. All Group I clones were identified as rDNA genes or fragments of chloroplast genome, whereas the majority of Group II clones showed significant homologies to retroelement sequences. Clones in Groups III and IV contained novel dispersed repeats with copy numbers 10^2–10^6/1C and two genus-specific tandem repeats. One of these belongs to the VicTR-B repeat family, and the other clone (S12) contains an amplified portion of the rDNA intergenic spacer. *In situ* hybridization using *V. sativa* metaphase chromosomes revealed the presence of the S12 sequences not only within rDNA genes, but also at several additional loci. The newly identified repeats, as well as the retroelement-like sequences, were characterized with respect to their abundance within individual genomes. Correlations between the repeat distributions and the current taxonomic classification of these species are discussed.

Abbreviations: C, DNA content of the unreplicated haploid nuclear genome; DAPI, 4′,6-diamidino-2-phenylindole; FISH, fluorescence *in situ* hybridization; *gag/pol*, retrotransposon open reading frame encoding the coat protein and enzyme components; IGS, intergenic spacer; *int*, integrase; LTR, long terminal repeat; ORF, open reading frame; PCR, polymerase chain reaction; PRINS, primed *in situ* labeling; rDNA, ribosomal DNA; RT, reverse transcriptase; SDS, sodium dodecyl sulfate; SSC, saline-sodium citrate buffer

Introduction

Repetitive DNA accounts for most of differences in genome size and genomic sequence composition observed between higher-plant species (Flavell *et al.*, 1974). Two main classes of repeats are distinguished, based on their genomic organization. Dispersed repeats are represented by various families of mobile elements (Kidwell and Lisch, 1997). Retrotransposons and retroelement-like sequences are particularly abundant in many plant genomes (Flavell *et al.*, 1992; Voytas *et al.*, 1992; SanMiguel and Bennetzen, 1998). and elements of the same type are present even in taxonomically distant species (Flavell *et al.*, 1997; Suoniemi *et al.*, 1998). Tandem repeats (satellite DNA) form arrays of various sizes, with their monomer units arranged in a head-to-tail orientation. The tandem repeats can be amplified to millions of copies, and form heterochromatic knobs and bands in nuclei and mitotic chromosomes, respectively (Schmidt and Heslop-Harrison, 1998). Different tan-

dem repeat sequences are preferentially amplified in a single species or a group of related species (Schmidt and Heslop-Harrison, 1993; King *et al.*, 1995; Maggini *et al.*, 1995; Nakajima *et al.*, 1996; Alix *et al.*, 1998; Nouzová *et al.*, 1999; Staginnus *et al.*, 1999).

Due to the tremendous variability in abundance and sequence composition of individual repeat families, a comprehensive analysis of the entire pool of repeats within a single species is very difficult. The same consideration applies to comparison of the repeats between different species, which therefore are usually focused on a few selected sequences (Flavell *et al.*, 1992; Schmidt and Heslop-Harrison, 1993; King *et al.*, 1995; Pearce *et al.*, 1996), or on comparisons of the total bulk of the repeats whilst neglecting the contribution of individual families (Chooi, 1971; Flavell *et al.*, 1974; Raina and Narayan, 1984). New experimental approaches are therefore needed to permit simultaneous analysis of large numbers of different repeats in multiple species. The recent advent of DNA microarray technologies may provide a tool for more efficient analysis of complex biological samples. Originally designed for monitoring of gene expression in man, yeast and higher plants (Schena *et al.*, 1995, 1996; Shalon *et al.*, 1996), DNA microarrays allow parallel processing of thousands of samples immobilized on a few square centimeters of a solid (usually glass) support (Kehoe *et al.*, 1999). In this paper, we describe the development of microarray-based methods for rapid screening and comparative analysis of repetitive sequences within selected *Vicia* species.

The genus *Vicia* includes 166 species (Allkin *et al.*, 1986), some of which, such as *V. faba* (broad bean), *V. narbonensis* (narbon vetch), and *V. sativa* (common vetch), are economically important crops. There is a considerable variation in chromosome size and number ($2n = 10$, 12, or 14), haploid nuclear DNA content (1.8–13.3 pg), proportion of repetitive DNA sequences, and abundance of individual repeats between different *Vicia* species (Chooi, 1971; Raina and Narayan, 1984; Maxted, 1995; Bennett *et al.*, 1998). This makes the genus ideal for research focused on the impact of amplification of individual types of repeats on the genome size, and on the molecular mechanisms of this amplification. Several studies of high-copy-number DNA sequences in *Vicia* spp. have been reported. Tandem repeats have been described that either are amplified only in one species (Maggini *et al.*, 1995; Nouzová *et al.*, 1999) or are present at high copy numbers in several related species (Macas *et al.*, 2000). The presence of dispersed repeats, usually related to retroelements, has also been described for several legume species (Kato *et al.*, 1985; Lehmann and Kozubek, 1993; Pearce *et al.*, 1996; Frediani *et al.*, 1999).

In this work, we describe the use of DNA microarrays as a tool for preliminary screening of novel repeats randomly cloned from *V. melanops*, *V. narbonensis*, and *V. sativa* genomes. Hybridization profiles were employed to characterize the abundance of the different repeat sequences across the genomes of 20 different legume species. This allowed classification of the sequences according to differences in their distribution patterns among related species, prior to further detailed analysis.

Materials and methods

Plant material

Seeds of field bean (*Vicia faba* ssp. *faba* var. *equina* Pers., $2n = 12$) cv. Inovec were obtained from Dr M. Vavák (Horná Streda, Slovakia). Seeds of the other Fabaceae species were received from the legume gene bank at Agritec, Šumperk, Czech Republic, or from IPK Gatersleben, Germany. Seeds of *Arabidopsis thaliana* L. ecotype Columbia were obtained from Dr. M. Ondřej (Institute of Plant Molecular Biology, Czech Republic). Seeds of *Vigna unguiculata* L. were provided by Prof. W.J. Broughton (University of Geneva, Switzerland).

Preparation of genomic DNA libraries

Total genomic DNA was extracted from leaves as described by Dellaporta *et al.* (1983). The DNA was digested with *Taq*I and ligated to a linker prepared by annealing oligonucleotides LKBX-D (5′-CGCATCCTCAGTCCGTAGCCATCACA-3′, 5′-phosporylated) and LKBX-K (5′-ATGGCTACGGACT GAGGATG-3′). The linker was in molar excess over the genomic DNA in order to saturate its free ends and prevent ligation of unrelated genomic DNA fragments. Purified fragments carrying the linker at both ends were then ligated into *Bst*XI-digested pcDNA2.1 vector (Invitrogen) and transformed into *Escherichia coli* strain XL-1 Blue MRF′ (Stratagene). Colonies obtained after *E. coli* transformation were used for colony hybridization with labeled genomic DNA, and the clones showing strong hybridization signals were selected for further analysis.

DNA microarrays

Inserts of the selected colonies were PCR-amplified directly from bacterial glycerol stocks with 5′ amino-modified M13-reverse and T7 primers and 2 μl of the stock per reaction. The reaction mix (50 μl volume) contained 1× PCR buffer, 2 mM $MgCl_2$, 0.2 mM dNTPs, 1.25 units of AmpliTaq-Gold polymerase (Perkin-Elmer) and 0.2 μM of each primer. The reaction was carried out in a 96 V Alpha unit of the PTC-200 DNA Engine (MJ Research) and involved 55 cycles of 94 °C for 1 min, 55 °C for 50 s, and 72 °C for 3 min. Cycling was preceded by an initial denaturation step (94 °C, 5 min) and followed by a final extension step (72 °C, 15 min). The amplification products were precipitated with isopropanol, and dissolved in 12 μl of 2× SSC. Control samples of the genomic DNAs were prepared for spotting by sonication (in order to break the high-molecular-weight DNA into 800–2500 bp fragments) and dilution in 2× SSC to final concentrations of 50 ng/μl and 200 ng/μl, respectively. Both types of probes (for 'probe' and 'target' nomenclature, see Phimister, 1999) were then spotted at 600 μm spacing on silane-coated slides (CEL Associates) using an adapted Biomek 2000 Laboratory Automation Workstation as described by Macas *et al.* (1998). The resultant microarrays contained 384 clones (arranged in four subarrays of 12 × 8 elements) from each of the three genomic libraries (*V. melanops*, *V. narbonensis*, and *V. sativa*), and a set of 96 control samples spotted in triplicate at different parts of the array. The control samples included genomic DNAs from all species used as hybridization targets (in 50 and 200 ng/μl dilutions), and various previously characterized repetitive sequences from *V. faba*. After microarray printing, the DNA was immobilized as described by Schena *et al.* (1996).

Fluorescent targets for microarray hybridizations were labeled by incorporation of Cy3- or Cy5-dUTP (Amersham) by using random priming or the polymerase chain reaction. The Vistra Random Prime Labeling Kit (Amersham) was used for labeling genomic DNAs and *V. faba* rDNA (clone Ver17) (Yakura and Tanifuji, 1983). Fragments of the *Ty1-copia* elements were amplified from *V. sativa* genome using the primers and reaction conditions described by Flavell *et al.* (1992). The fragment of pcDNA2.1 comprising the multiple cloning site devoid of insert was PCR-amplified with M13 reverse and T7 primers. Microarrays were denatured by submersion in deionized water for 2 min at 92–95 °C followed by a 15 s incubation in 96% ethanol and air drying. Hybridization mix (18 μl) containing Cy3- and Cy5-labeled targets, 4× SSC, 0.09% SDS and 20 μg herring sperm DNA was denatured for 2.5 min at 100 °C, cooled on ice and applied onto the microarray slides preheated to 65 °C. The arrays were covered with 45 mm × 22 mm coverslips and incubated overnight in a humid chamber equilibrated to 60 °C. Post-hybridization washes were performed at 30 °C in 5× SSC, 0.1% SDS (5 min), 0.2× SSC, 0.1% SDS (5 min), and 0.2× SSC (2 min). The slides were air-dried and scanned with a ScanArray 3000 microarray scanner (GSI-Lumonics).

In all experiments, a Cy3-labeled target (either one of the genomic DNAs, rDNA or *Ty1-copia* fragments) was used together with Cy5-labeled fragment of the pcDNA2.1 vector. Since all clones immobilized on the arrays contained the same portion of this plasmid, the Cy5 signal intensity provided a measure of the amounts of DNA present in individual spots, and was used for equalization of Cy3 signals as follows. Two images were always collected by separately scanning the given array using Cy3- and Cy5-specific filter sets (Figure 1A, B). The images were imported into the Adobe Photoshop 5.0 program, converted to 8-bit image mode and maximum imput levels of the Cy3 scans were set to 100. Then, the two images were overlayed using a 'Multiply' blending function with Cy3 image set as Source 1 and inverted (negative) Cy5 image as Source 2. As a consequence, Cy3 intensities were changed inversely proportionally to the corresponding Cy5 signals (in other words, Cy3 signals were reduced in spots containing large amounts of amplified fragments resulting in high Cy5 signals, and vice versa). Finally, the resulting gray-scale images were pseudocolored using a red-green-blue-gray indexed color scheme, such that the highest signals were red and the lowest were black (Figure 1C).

Dot-blot and Southern hybridization

Membrane hybridizations were done with the AlkPhos Direct labeling and detection kit with the DNA immobilized on Hybond N+ membranes (Amersham). The inserts selected as hybridization probes were PCR amplified from the plasmid with M13-forward and reverse primers and digested with *Taq*I to remove polylinker sequences. The fragments were gel-purified prior to labeling. Hybridizations were performed according to manufacturer's recommendations at stringent temperatures (60 °C) and the signals were detected with either the chemiluminescent substrate CDP

Table 1. Taxonomy of Fabaceae species used in this study.

Tribe[a]	Genus	Section[b]	Species	Abbreviation	1C[c] [pg]	2*n*[c]
Vicieae	*Vicia*	Hypechusa	*V. lutea* L.	Vlut	7.40	14
			V. melanops Sibth. & Smith	VM	10.00	10
			V. hybrida L.	VH	6.78	12
			V. pannonica Grantz	Vpan	6.75	12
		Cracca	*V. villosa* Roth	VV	2.28	14
		Attosa	*V. sepium* L.	Vsep	4.68	14
		Vicia	*V. grandiflora* Scop.	VG	3.35	14
			V. sativa L.	VS	2.25	12
		Wiggersia	*V. lathyroides* L.	Vlath	2.63	10
		Narbonensis	*V. narbonensis* L.	VN	7.28	14
		Faba	*V. faba* L.	VF	13.50	12
		Peregrina	*V. peregrina* L.	Vper	9.48	14
			V. michauxii Sprengel	Vmich	8.30	14
	Pisum		*P. sativum* L.	PS	4.65	14
			P. elatius (M.B.) Stev.	PE	4.43	14
Cicereae	*Cicer*		*C. arietinum* L.	CA	0.95	16
Phaseoleae	*Glycine*		*G. max* (L.) Merr	GM	1.13	40
	Phaseolus		*P. vulgaris* L.	PV	0.60	22
	Vigna		*V. unguiculata* (L.) Walp.	VU	0.60	22
Genisteae	*Lupinus*		*L. angustifolius* L.	LA	0.93	26

[a]according to Polhill and Raven (1981).
[b]according to Kupicha (1976) and Maxted (1995).
[c] according to Bennett et. al. (1998).

Star (exposure on a X-ray film) or the chemifluorescent substrate ECF (detected with a Storm scanner, Molecular Dynamics).

For confirmation of the results obtained in microarray hybridizations, 3 ng of PCR-amplified inserts were dot-blotted together with three dilutions (100, 10 and 1 ng) of genomic DNAs of twenty Fabaceae species (see Table 1) and *Arabidopsis thaliana*, and sequentially hybridized to labeled genomic DNAs of the same species. To make results of individual hybridizations comparable, the signals were quantified with the Storm scanner, and were expressed as a percentage of the signal present in a dot containing 1 ng of genomic DNA from the same species as the one used as a target.

For rough estimation of copy numbers of repeats from the Groups III and IV in tested Fabaceae genomes, amounts corresponding to 10^5 copies of the haploid genomes were dot-blotted in duplicate along with 10^7–10^{11} copies of respective repetitive sequences that were also used as hybridization probes. Copy numbers were then estimated by comparing signal intensities of genomic DNAs and the dilution series of a given clone. Calculations of the percentage of genomes occupied by individual sequences were based on length, estimated copy numbers of individual probes, and the assumption that 1 pg of genomic DNA equals 9.65×10^8 bp (Bennett and Smith, 1976).

For Southern hybridizations, 2 μg of digested genomic DNAs were run on 1.5% agarose gels and blotted onto the membranes by capillary transfer.

DNA sequencing and sequence analysis

Clones selected for sequencing were subcloned into pBSK+II vector (Stratagene) and sequenced by the dideoxy chain termination method (Sanger *et al.*, 1977) with M13-forward and reverse primers. The resultant sequences were searched for homologies in GenBank (release 115.0) and EMBL (release 61) databases with BLASTN or BLASTX 2.0.2 (Altschul *et al.*, 1997) or FASTA 3.2 (Pearson and Lipman, 1988). The homologies described as significant in this paper have maximal expectation values of at most 10^{-5}. Pairwise comparisons between the sequences as well as other sequence analyses (searches for direct and inverted repeats, secondary structure analysis) were performed with PC/Gene 6.60 (IntelliGenetics) and Dot-plot 3.0 (Ramin Nakisa, Oxford University).

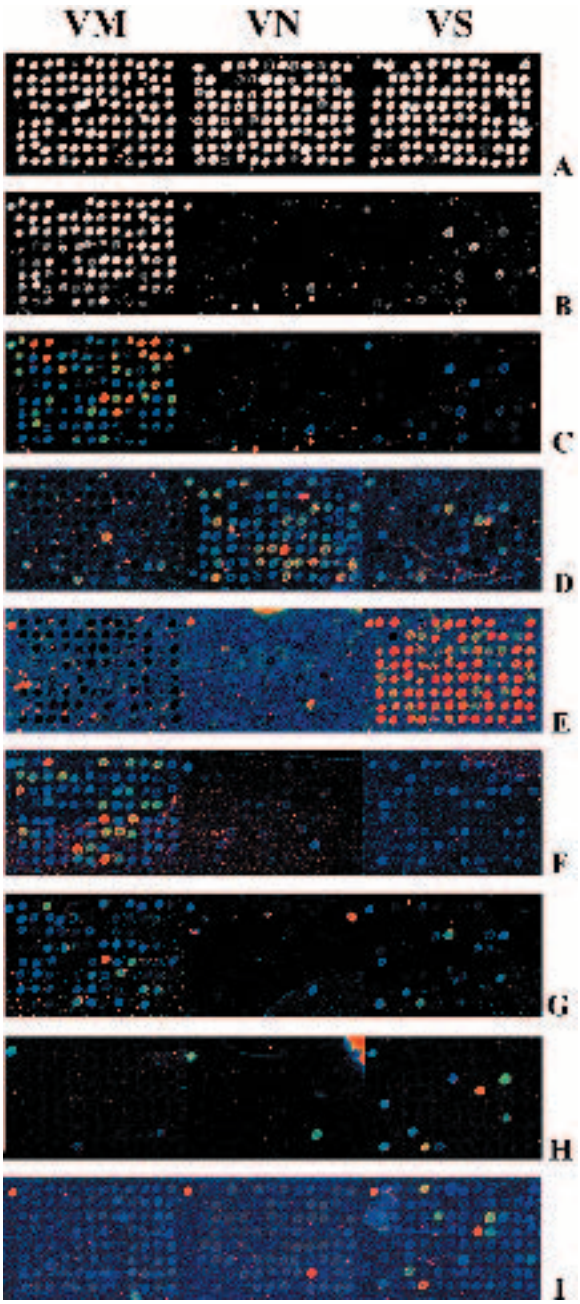


Figure 1. DNA microarray hybridization. Sequences from genomic DNA libraries of *Vicia melanops*, *V. narbonensis*, and *V. sativa* were amplified by PCR using primers derived from the cloning vector, and were spotted on slides. Each slide was hybridized simultaneously with Cy5-labeled fragment of the cloning vector to estimate amounts of the spotted DNA (A) and with Cy3-labeled total genomic DNA or a sequence-specific target (B). Merging these two gray-scale images resulted in a pseudocolored picture reflecting hybridization intensities of individual sequences (C). Panels C to I show part of the arrays after hybridization with the following targets: *V. melanops* (C), *V. narbonensis* (D), *V. sativa* (E), *V. hybrida* (F), *V. pannonica* (G), *Vigna unguiculata* (H) and rDNA (I). Each image illustrates the same segment of the microarrays, containing 96 clones each from *V. melanops* (VM), *V. narbonensis* (VN), and *V. sativa* (VS).

Fluorescence in situ *hybridization (FISH)*

The experiments were performed using purified suspensions of *V. sativa* metaphase chromosomes dried onto microscope slides. The suspensions were prepared according to Gualberti *et al.* (1996) with synchronized root tip meristems (Macas *et al.*, 2000). The probes were labeled with biotin-dUTP (Boehringer-Mannheim) in a PCR reaction using T3 and T7 primers and the corresponding plasmid clone as a template. Denaturation of chromosomes was carried out for 3 min at 94 °C in 1× PCR buffer containing 4 mM $MgCl_2$ and was followed by slide dehydration using an ethanol series. Hybridization was performed overnight at 37 °C in a mix comprising 2× SSC, 50% formamide, 10% dextran sulfate, 125 ng/μl of sheared calf thymus DNA, 0.125% SDS, and 0.025 ng/μl of labeled probe. Post-hybridization washes included 2× SSC at 42 °C for 5 min followed by stringent washing in 50% formamide/2× SSC at 42 °C for 10 min. Probe detection was done using the Tyramide Signal Amplification Indirect system (NEN Life Sciences Products) with streptavidin-fluorescein at the final step. Chromosomes were counterstained with DAPI (4′,6-diamidino-2-phenylindole) and examined under a Nikon Eclipse 600 microscope equipped with appropriate filter sets. The images were acquired with a CCD camera and Lucia software (Laboratory Imaging). Individual types of *V. sativa* chromosomes were identified based on their morphology and characteristic DAPI banding.

Results

Three *Vicia* species (*V. sativa*, *V. melanops*, and *V. narbonensis*) differing in genome size, chromosome numbers and assignment to taxonomic sections within the genus (Table 1) were used for preparation of genomic DNA libraries. About 3000 clones from each library were screened for repetitive sequences by hybridization with labeled genomic DNA of the species used for library construction. Considering that random fragments were cloned in the libraries (as confirmed by analysis of selected clones described below) and that the average insert size was 563 bp, screening 3000 clones should detect all repeats comprising at least 0.1% of the genome with a 95% probability (Ausubel *et al.*, 1991). The clones showing strong hybridization signals (384 from each library) were selected for microarray preparation.

The microarrays containing 1440 sample spots (1152 clones from the libraries and 96 control samples spotted in triplicate) were hybridized with genomic DNA targets of 17 legume species. These included thirteen *Vicia* species representing eight different taxonomic sections, together with pea (*Pisum sativum*), soybean (*Glycine max*), common bean (*Phaseolus vulgaris*), and cowpea (*Vigna unguiculata*) (Table 1). Hybridization was also performed with labeled Ver17 and *rt* to identify clones containing rDNA and retroelement reverse transcriptase sequences, respectively. As expected, targets derived from the species used for library constructions produced hybridization signals on the majority of clones from the corresponding library (Figure 1C–E). The signals were also more prominent when the species from the same section were used as targets (Figure 1F, G) and weak or missing with targets from evolutionary distant species (Figure 1H). Clones showing signals with the latter were often found to be rDNA sequences (Figure 1I). These were excluded from further analysis.

Based on hybridization signals with individual targets, the clones were divided into ten groups: (1) clones hybridizing with all tested species (14 clones), (2) clones hybridizing with almost all tested *Vicia* species and with *Pisum sativum* (139 clones), (3) clones hybridizing with almost all tested *Vicia* species but not with *Pisum sativum* (55 clones), (4–9) six groups of clones giving signals only with different subsets of *Vicia* species (470 clones), and (10) clones giving a strong signal only with a single species (the one used for the library construction) (53 clones). Nine to fourteen clones representing individual groups were selected for further analysis. The DNA of these clones (111 in total) was dot-blotted onto nylon membranes and hybridized with the same set of targets as used in the microarray experiments. Comparison of the results obtained using this conventional technique with the microarray data revealed that about 80% clones gave the same hybridization profiles and that these clones fall into similar groups (data not shown). However, it was realized that due to minor discrepancies in signal intensities the only reliable classification is limited to four basic groups, comprising (I) clones hybridizing with all tested species, (II) clones hybridizing with most of *Vicia* and *Pisum* species, (III) clones hybridizing to a limited number of *Vicia* species, and (IV) clones showing preferential hybridization to a single *Vicia* species. A total of 65 clones were then selected to evenly represent all four groups and sequenced. Insert lengths, GenBank accession numbers, and hybridization profiles of these clones are given in Table 2.

Groups I and II

All Group I clones (characterized by hybridization to all tested species) showed significant homology with conserved high copy number sequences from nuclear (rDNA) or chloroplast genomes (Table 3).

The Group II clones contained repeats of retroelement origin. Of 26 clones, 20 (77%) were directly identified as retroelement-like sequences based on their homologies to conserved reverse transcriptase (*rt*) or integrase (*int*) regions of known retroelements (Table 3). The remaining six clones (M4, N40, S2, S5, N16, N30) gave the best homology scores with *V. faba* and *P. sativum* repeats of unknown origin. However, the clone N16 also displayed homology to a *Medicago sativa* transcribed middle repetitive element RPH8 (GB:L39961) and the clone N30 to a *pol* region of Calypso retrovirus-like element from *Glycine max* (GenBank AF186186). Moreover, clones M4, N40, S2 and S5 share sequence overlaps with the clones M1 and M38 that contain regions of homology to the integrase, suggesting that they might also be classified as retroelement-related sequences.

Sequence analysis revealed the presence of internal subrepeats of 35 bp in clones M1, M38, N40, S2, and S5. The subrepeats are present in 1 to 3 copies in individual clones and their mutual homologies range between 58% and 97%. In clones containing homology region to the integrase sequence (M1, M38) the subrepeats are located at the 3′ direction from that region (data not shown).

Groups III and IV

The majority of the clones from these groups represent newly described repetitive elements having no detectable homology to known sequences. For this reason they were subjected to more detailed analysis to obtain data concerning their genomic organization and abundance in different *Vicia* species (Table 4). Southern blot hybridizations to digested genomic DNAs of various species revealed that most of the clones produced patterns typical for dispersed repeats (smears with several prominent bands lacking regular spacing, Figure 2). That also included clones containing two or three copies of internal 33–72 bp subrepeats (M6, S11, N14, N38, N9, N37, N3, N21).

Three clones (N6, S38, and S12) contained subrepeats arranged in a tandem manner lacking any

Table 2. Assignment of the sequenced clones to groups based on their abundance in individual Fabaceae species.

Group	Clone	GenBank accession number	Length (bp)	Genomic probe																			
				V lut	V M *	V H	V pan	V V	V sep	V G	V S *	V lath	V N *	V F	V per	V mich	P S	P E	C A	G M	P V	V U	L A
I	M27	AJ391748	509	2	2	2	2	2	3	2	2	2	2	2	3	2	2	5	2	3	3	4	1
	M29	AJ391749	675	2	2	2	2	3	3	2	2	2	3	2	3	3	3	5	3	4	3	4	2
	S9	AJ391750	344	2	2	2	2	4	4	2	2	2	2	2	3	3	3	5	4	4	5	5	1
	S17	AJ391751	277	2	2	2	2	3	3	2	2	2	2	2	2	2	2	5	2	3	4	4	1
	S19	AJ391752	344	2	2	2	2	4	4	2	2	2	2	2	3	3	3	5	5	4	5	5	2
	S35	AJ391753	324	2	2	2	1	3	2	2	2	2	2	2	2	2	2	4	2	4	4	4	1
II	M1	AJ391775	449	3	5	5	4	5	5	3	2	3	3	2	4	4	4	5	0	0	0	0	0
	M2	AJ391759	757	2	5	5	5	4	5	3	2	2	2	2	2	0	0	0	0	0	0	0	0
	M4	AJ391779	821	5	5	5	2	5	5	3	2	4	3	3	5	5	3	5	0	0	0	0	0
	M8	AJ391761	504	2	5	5	5	5	5	4	2	0	3	2	0	2	0	0	0	0	0	0	0
	M9	AJ391760	580	2	5	5	5	4	5	3	2	0	3	2	0	2	0	3	0	0	0	0	0
	M18	AJ391754	382	2	5	4	5	3	5	3	2	2	3	2	2	0	0	0	0	0	0	0	0
	M19	AJ391764	642	3	5	5	5	4	5	3	2	2	3	2	3	2	3	5	0	0	0	0	0
	M37	AJ391765	642	2	5	5	5	5	5	3	2	3	3	1	3	3	4	5	0	0	0	0	0
	M38	AJ391774	718	5	5	5	4	5	5	3	2	5	3	2	5	5	4	5	0	0	0	0	0
	M43	AJ391755	490	2	5	3	4	3	4	2	2	2	2	1	0	0	0	0	0	0	0	0	0
	N13	AJ391756	474	4	2	2	1	3	4	2	2	4	5	2	3	3	2	4	0	0	0	0	0
	N16	AJ391771	353	2	0	0	0	0	2	2	0	0	4	0	2	0	0	5	0	0	0	0	0
	N20	AJ391757	663	4	3	2	1	2	5	3	2	5	5	2	4	4	2	4	0	0	0	0	0
	N30	AJ391780	409	3	0	0	1	3	5	3	2	4	5	2	2	2	3	5	0	0	0	0	0
	N32	AJ391768	379	2	2	2	1	2	3	2	2	3	3	2	2	2	0	0	0	0	0	0	0
	N35	AJ391770	396	4	2	3	0	2	3	2	0	0	4	2	0	3	0	4	0	0	0	0	0
	N40	AJ391777	540	4	3	3	2	2	5	3	2	4	4	3	5	5	3	5	0	0	0	0	0
	S2	AJ391778	211	3	2	3	1	2	4	3	2	4	3	3	5	5	3	4	0	0	0	0	0
	S5	AJ391776	704	3	2	2	1	2	4	2	3	4	3	2	4	4	2	4	0	0	0	0	0
	S7	AJ391773	385	2	2	3	0	3	3	2	3	2	3	2	2	2	2	4	0	0	0	0	0
	S15	AJ391762	500	4	2	2	1	2	5	3	3	4	3	2	3	3	2	3	0	0	0	0	0
	S23	AJ391763	277	2	0	0	0	2	2	2	2	2	3	2	2	2	0	0	0	0	0	0	0
	S28	AJ391767	494	2	2	2	1	2	4	2	3	4	2	2	3	2	0	0	0	0	0	0	0
	S33	AJ391769	339	3	4	2	1	2	3	2	3	4	3	2	3	3	2	0	0	0	0	0	0
	S34	AJ391766	420	2	5	4	3	4	5	3	2	2	3	2	2	3	2	5	0	0	0	0	0
	S36	AJ391758	533	3	2	0	1	2	4	3	4	5	3	2	3	2	2	0	0	0	0	0	0
III	M6	AJ391781	1061	2	5	5	4	2	5	2	2	0	2	0	0	0	0	0	0	0	0	0	0
	M12	AJ391800	988	0	5	2	2	0	2	0	0	0	2	0	2	0	0	0	0	0	0	0	0
	M20	AJ391797	489	0	4	2	2	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0
	M23	AJ391795	537	2	5	0	2	0	2	0	0	0	2	0	2	0	0	0	0	0	0	2	0
	M25	AJ391789	506	0	4	2	2	0	2	1	0	0	3	2	0	0	0	0	0	0	0	0	0
	M26	AJ391791	643	0	4	0	2	0	2	0	0	0	2	0	1	0	0	0	0	0	0	0	0
	M32	AJ391792	562	0	5	2	3	0	2	0	0	0	2	0	2	0	0	0	0	0	0	0	0
	M34	AJ391798	306	0	3	3	2	0	2	0	0	0	2	1	0	0	0	0	0	0	0	0	0
	M36	AJ391793	222	0	5	2	2	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0
	M39	AJ391794	337	0	5	0	3	0	2	0	0	0	2	2	0	0	0	0	0	0	0	0	0
	M41	AJ391799	332	0	5	3	3	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0

Table 2 continued.

Group	Clone	GenBank accession number	Length (bp)	Genomic probe																			
				V lut	V M *	V H	V pan	V V	V sep	V G	V S *	V lath	V N *	V F	V per	V mich	P S	P E	C A	G M	P V	V U	L A
	N5	AJ391784	348	2	0	2	1	0	2	0	0	0	5	0	2	0	0	0	0	0	0	0	0
	N8	AJ391785	355	2	0	2	0	2	2	0	0	0	5	0	0	0	0	0	0	0	0	0	0
	N12	AJ391786	358	2	2	2	2	2	2	0	0	0	5	0	0	0	0	0	0	0	0	0	0
	N28	AJ391783	357	2	2	2	2	2	3	0	0	0	5	0	0	0	0	0	0	0	0	0	0
	N33	AJ391803	333	0	0	0	0	2	2	0	0	2	3	0	2	0	0	0	0	0	0	0	0
	N38	AJ391796	401	2	0	2	2	2	2	0	0	0	4	1	0	0	0	0	0	0	0	0	0
	S1	AJ391804	559	0	0	0	0	0	2	1	2	0	2	0	0	0	0	0	0	0	0	0	0
	S8	AJ391772	289	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0
	S11	AJ391782	1040	2	5	3	2	3	5	3	3	0	2	0	0	0	0	0	0	0	0	0	0
	S12	AJ391788	352	2	0	2	0	0	2	2	5	0	2	0	0	0	0	0	0	0	0	0	0
	S13	AJ391790	385	0	0	0	0	0	2	0	2	0	2	2	0	0	0	0	0	0	0	0	0
	S26	AJ391805	506	0	0	0	0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0
	S38	AF191788	308	2	2	2	0	0	5	5	4	0	2	1	0	0	0	0	0	0	0	0	0
IV	M16	AJ391801	426	0	3	0	1	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0
	M21	AJ391802	211	0	5	0	0	0	2	0	0	0	2	2	0	0	0	0	0	0	0	0	0
	N3	AJ391809	656	0	0	0	0	2	2	0	0	0	5	0	0	0	0	0	0	0	0	0	0
	N6	AJ391811	520	0	0	0	0	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0	0
	N9	AJ391807	322	2	0	0	0	0	2	0	0	0	5	0	0	0	0	0	0	0	0	0	0
	N14	AJ391787	688	0	0	0	0	0	2	0	0	0	5	0	0	0	0	0	0	0	0	0	0
	N15	AJ391806	550	0	0	0	0	0	2	0	0	0	4	0	0	0	0	0	0	0	0	0	0
	N21	AJ391810	418	0	0	0	0	2	2	0	0	0	5	2	0	0	0	0	0	0	0	0	0
	N37	AJ391808	317	0	0	0	0	0	2	1	0	0	4	2	0	0	0	0	0	0	0	0	0

The clones were sorted into four groups based on dot-blot hybridizations with twenty Fabaceae species (for abbreviations of the species names, see Table 1). I: clones hybridizing with all tested species; II: clones hybridizing with most of the *Vicia* and *Pisum* species; III: clones hybridizing only with a subset of *Vicia* species; IV: clones hybridizing mainly with single *Vicia* species used for library preparation. Signal intensities after hybridization with individual genomic probes are expressed in a scale from 0 (no signal) to 5 (the highest signal). Three *Vicia* species used for clone isolations are marked by asterisks; names of the clones indicate a species of origin (M, *V. melanops*; N, *V. narbonensis*; S, *V. sativa*) and the clone serial number. Signal intensities are highlighted: 1–2, yellow shading; 3–5, green shading.

surrounding sequences and produced ladder-like patterns on Southern blots (Figure 3A, B). The clone N6 comprised nine 66 bp subrepeats; however, its hybridization pattern was not regular (Figure 3A) suggesting the sequence heterogeneity or irregular arrangement in *V. narbonensis* genome. In contrast, the two other clones produced patterns typical of tandemly organized satellite DNAs. Clone S38 containing eight 38 bp subrepeats was identified as a member of previously described VicTR-B repeat family (Macas *et al.*, 2000). Clone S12 (Figure 3B) comprising two 174 bp monomers showed sequence homology to tandem subrepeats within 25S/18S rDNA intergenic spacers of *V. angustifolia* (GenBank X61082, 89% homology in 111 bp overlap), *V. faba* (GenBank X16615, 66% homology in 174 bp overlap), and other legume species.

Southern blots of *V. sativa* genomic DNA probed with S12 displayed band ladders with spacing corresponding to the repeat monomer (174 bp) ranging up to multimers of several kilobases in size (Figure 3B). Although the S12 sequences were also detected in other *Vicia* species, none of them produced the same hybridization pattern as seen for *V. sativa* (Figure 3C). Moreover, the copy number of S12 in *V. sativa* was at least an order of magnitude higher (10^4–10^5) than for the other species ($<10^4$, Table 4). These data suggested that in the *V. sativa* genome the S12 sequences might be amplified as satellite DNA located outside the rDNA loci. To test this hypothesis *in*

Table 3. Characterization of clones from Groups I and II.

Group	Clone[a]	Protein homology	DNA homology				Homologies with the other newly isolated clones
			sequence name	sequence type	organism	GenBank accession number	
I.	M27	RNA polymerase	chloroplast DNA	gene	*Pisum sativum*	X03912	–
	M29	NADH oxidoreductase	chloroplast DNA	gene	*Nicotiana tabacum*	Z00044	–
	S9	–	18S rRNA	gene family	*Pisum sativum*	U43011	S12
	S19	–	18S rRNA	gene family	*Pisum sativum*	U43011	S9
	S17	–	18S rRNA	gene family	*Trifolium repens*	Af071069	–
	S35	–	chloroplast 16S rRNA	gene	*Glycine max*	M37149	–
II.	M2*		P2	repetitive element	*Pisum sativum*	AJ002213	M9, M8, M18, M43, N13, S15, S36
	M8		BamH 1750	repetitive element	*Vicia faba*	M16858	M2, M9, N13, S15
	M9		PIS1-1	repetitive element	*Pisum sativum*	AF155741	M2, M8, N13, S15
	M18*	putative reverse	PIS1-5	repetitive element	*Pisum sativum*	AF155745	M2, M43, N13, N20, S36
	M43*	transcriptases of	PIS1-5	repetitive element	*Pisum sativum*	AF155745	M2, M18, N13, N20, S36
	N13*	Ty3/*gypsy* type	BamHI 990	repetitive element	*Vicia faba*	M16855	M2, M8, M9, M18, M43, N20, S15, S36
	N20*	retroelements from	BamHI 990	repetitive element	*Vicia faba*	M16855	M18, M43, N13, S36
	S15	different organisms	BamH 1750	repetitive element	*Vicia faba*	M16858	M2, M8, M9, N13
	S36*		BamHI 990	repetitive element	*Vicia faba*	M16855	M2, M18, M43, N13, N20
	S23		BamHI 900	repetitive element	*Vicia faba*	M16854	–
	M19		BamHI 1500	repetitive element	*Vicia faba*	M16857	M37, N32, S28, S34
	M37		BamHI 1500	repetitive element	*Vicia faba*	M16857	M19, N32, S28, S34
	N32		BamHI 1500	repetitive element	*Vicia faba*	M16857	M19, M37, S33
	N35	putative integrase from	BamHI 1500	repetitive element	*Vicia faba*	M16857	N32, S7, S33
	S34	*Nicotiana tabacum*	BamHI 1500	repetitive element	*Vicia faba*	M16857	M19, M37, S28
	S28		BamHI 1500	repetitive element	*Vicia faba*	M16857	M19, M37, S34
	S33		BamHI 1500	repetitive element	*Vicia faba*	M16857	N32, N35
	S7		PIS2-14	repetitive element	*Pisum sativum*	AF155753	N35
	M1		PIS2-33	repetitive element	*Pisum sativum*	AF155770	M38, N40, S2
	M38		PIS2-33	repetitive element	*Pisum sativum*	AF155770	M1
	M4	–	TIV10	repetitive element	*Vicia faba*	AJ222868	M38
	N40	–	PIS2-13	repetitive element	*Pisum sativum*	AF155752	M1, S2
	S2	–	PIS2-13	repetitive element	*Pisum sativum*	AF155752	M1, N40, S5
	S5	–	PIS2-13	repetitive element	*Pisum sativum*	AF155752	S2
	N16	–	BamHI 1150	repetitive element	*Vicia faba*	M16856	–
	N30	–	PIS7	repetitive element	*Pisum sativum*	AF155747	–

[a]clones hybridizing with *rt* probe are marked by asterisks.

situ hybridization experiments were performed with metaphase *V. sativa* chromosomes and rDNA and S12 probes (Figure 4). The rDNA probe labeled secondary constrictions of two pairs of acrocentric chromosomes (Figure 4A). The S12 probe also labeled the rDNA loci, but produced additional signals on 3 acrocentric chromosome pairs (Figure 4B), thus confirming the location of these sequences outside the rDNA genes.

Table 4. Characterization of clones from Groups III and IV.

Group	Clone	V lut	V M *	V H	V pan	V sep	V G	V S *	V lath	V N *	V F	V per	V mich	P S	P E	Proportion of original genome (%)	DNA homologies[a] (% identity; overlap bp)
III	M6	2–3	5–6	4–5	5–6	2–3	3–4	3–4	3–4	–	2–3	–	–	–	–	1–10	S11
	M12	–	4–5	3–4	4–5	–	–	–	–	–	–	–	–	–	–	0.1–1	–
	M20	2	5–6	3–4	4–5	–	–	2	–	–	–	–	–	–	–	0.5–5	–
	M23	3–4	4–5	3–4	3–4	2–3	2–3	3–4	–	2	3–4	2–3	2–3	2–3	3–4	0.05–0.5	–
	M25	2–3	4	3–4	3–4	–	3–4	3–4	2–3	2–3	3–4	3–4	3–4	3–4	3	0.05	S13
	M26	3–4	4–5	3–4	4	2	3	2–3	2	3–4	2	3–4	3–4	3–4	3	0.07–0.7	***PS-GS3B*** (69; 316), ***PS-rcpme1*** (55; 328)
	M32	2–3	5–6	4	5	2–3	3–4	2–3	2–3	–	3–4	2	–	–	–	0.6–6	***PS-rcpme 1***(63; 189)
	M34	3	4	4	3–4	2	2	–	–	–	3	2–3	2–3	2	–	0.03	–
	M36	3	4–5	4	4–5	2	3–4	3–4	3	–	3–4	–	–	–	–	0.02–0.2	***PS-rcpme1*** (60; 224), ***PIS25*** (62; 165), ***MT BAC 1*** (66; 119)
	M39	2–3	4–5	3–4	4–5	2	2	2	–	3	–	2–3	–	2	2	0.03–0.3	PS-rcpme1 ***(69; 332)***, MT BAC 2 ***(64; 259)***, MS-RPE15 ***(60; 289)***
	M41	2–3	5–6	3–4	5–6	–	–	2–3	–	–	2–3	–	–	–	–	0.3–3	–
	N5	3–4	2–3	3–4	3	2–3	–	–	–	2–3	4–5	2–3	2–3	2–3	–	0.05–0.5	N8, N28, N14, N12
	N12	3–4	3	3–4	3	2–3	–	–	–	3	5–6	2–3	2–3	2–3	–	0.5–5.0	N28, N14, N5, N8
	N33	–	2	–	–	–	–	–	–	–	4–5	2	2	–	–	0.05–0.5	–
	N38	3–4	2–3	3–4	3	2–3	–	–	–	–	4–5	–	2	–	–	0.06–0.6	N14
	S1	2–3	3	3–4	2–3	2–3	3–4	3	4	2	2–3	2–3	2–3	2	2–3	0.2	PIS4 ***(65; 315)***
	S8	2–3	3	2–3	2–3	–	3–4	3–4	4–5	3–4	3–4	3	3	3	2–3	0.1–1	BamHI 1150 ***(66; 137)***
	S11	3	4–5	3–4	4–5	3	4–5	4–5	4–5	–	3	–	–	–	–	0.5–5	M6
	S12	3–4	–	–	–	–	3–4	3–4	4–5	–	2–3	3–4	2	–	–	0.2–2	IGS subrepeat ***(89; 111)***
	S13	2–3	2–3	2–3	2–3	2	3–4	2–3	4–5	3–4	3–4	2–3	2–3	2–3	2–3	0.2–2	M25
	S26	–	–	–	–	–	–	–	2–3	–	–	–	–	–	–	0.002–0.02	–
	S38	3–4	3–4	–	–	5–6	6	6–7	3–4	5–6	2–3	–	–	–	–	25	VicTR-B ***(90; 38)***

Table 4 continued.

Group	Clone	V lut	V M *	V H	V pan	V sep	V G	V S *	V lath	V N *	V F	V per	V mich	P S	P E	Proportion of original genome (%)	DNA homologies[a] (% identity; overlap bp)
IV	M16	2–3	4–5	3–4	3–4	–	–	–	–	–	3–4	2–3	–	–	–	0.04–0.4	–
	M21	–	4–5	2–3	3–4	–	–	–	–	–	–	–	–	–	–	0.02–0.2	–
	N3	–	–	–	–	2–3	2–3	2–3	–	–	6	3–4	–	–	–	9	N21
	N6	–	2	2–3	2	–	–	–	–	–	5–6	2	–	–	–	0.7–7.0	–
	N9	3–4	2–3	2–3	2–3	–	–	–	–	2–3	5–6	2	–	–	2–3	0.9–9	N37
	N14	2–3	–	–	–	–	–	–	–	2	5–6	–	–	–	–	0.9–9	N28, N5, N8, N12, N38
	N15	–	–	–	–	–	–	–	–	–	4–5	–	–	–	–	0.08–0.8	–
	N21	–	–	–	–	2–3	2–3	2–3	–	2	5–6	3–4	–	–	–	0.6–6	N3
	N37	3–4	2–3	2–3	2–3	–	2	3–4	2–3	2–3	5	2	–	–	2	0.4	N9

Abundance of sequences in an original genome and in genomes of other *Vicia* and *Pisum* species is given as copy numbers per haploid genome (1*C*). A 5 corresponds to 10^5, 4 to 10^4, 3 to 10^3 and 2 to 10^2 copies/1*C*, respectively; – indicates copy number less than 10^2/1*C* (copy numbers are also highlighted: 10^3–10^4/1C, yellow shading; 10^4/1C and higher, green shading). *Vicia* species used for cloning are marked by asterisks, abbreviations of the species names are given in Table 1.

[a]Homologies to other newly isolated clones are indicated by the clone names. Homologies to GenBank entries are indicated in bold italics and correspond to: ***PS GS3B***, sequence upstream from *Pisum sativum* pectin methylesterase ORF *(*Af081457); ***PS rcpme1***, sequence upstream from *Pisum sativum* cytosolic glutamine synthetase ORF (U28925); ***PIS25***, dispersed repeat from *Pisum sativum* (AF155762); ***MT BAC 1***, BAC clone 68P17 from *Medicago truncatula* (AQ841326); ***MT BAC 2***, BAC clone 37B11 from *Medicago truncatula* (AQ917489); ***MS-RPE15***, middle repetitive DNA from *Medicago sativa* (L39960); ***PIS4***, dispersed repeat from *Pisum sativum* (AF155744). ***BamHI 1150***, family of dispersed DNA repeats from *Vicia faba* (M16856); ***IGS subrepeat***, 25S/18S rDNA intergenic spacer tandem subrepeat from *Vicia angustifolia* (X61082); ***VicTR-B***, consensus sequence of *V. sativa* tandem repeat family (Macas *et al.*, 2000); average homology of 8 monomer units of clone S38 to the consensus *VicTR-B* sequence is given.

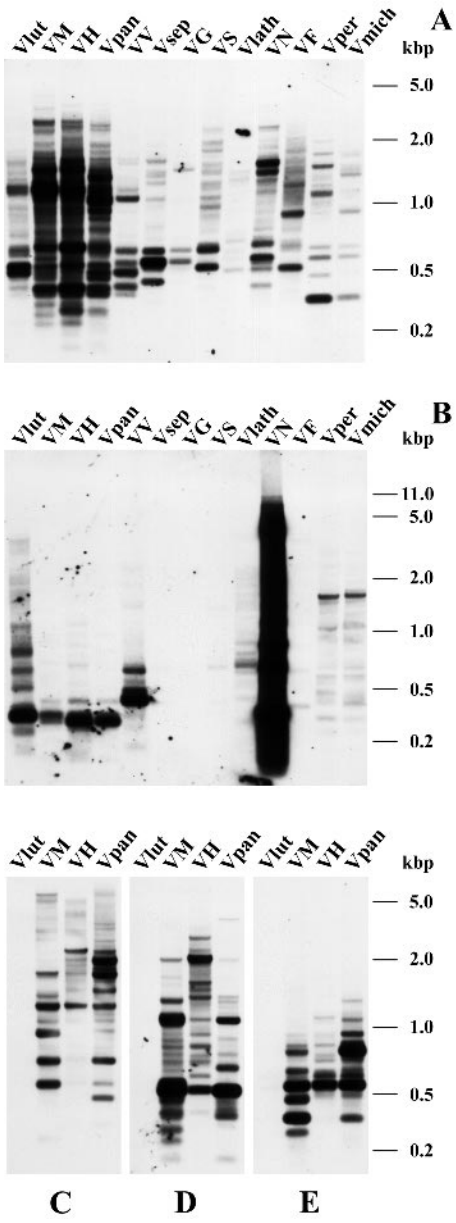


Figure 2. Organization of repeats M34, N5, M12, M20 and M41 in the genomes of *Vicia* species. Southern blots of genomic DNAs from different *Vicia* species digested with *Taq*I were hybridized with labeled sequences M34 (A) and N15 (B) (exposure time 17 h) and with M12 (C), M20 (D) and M41 (E) (exposure time 10 min). For abbreviations of species names, see Table 1.

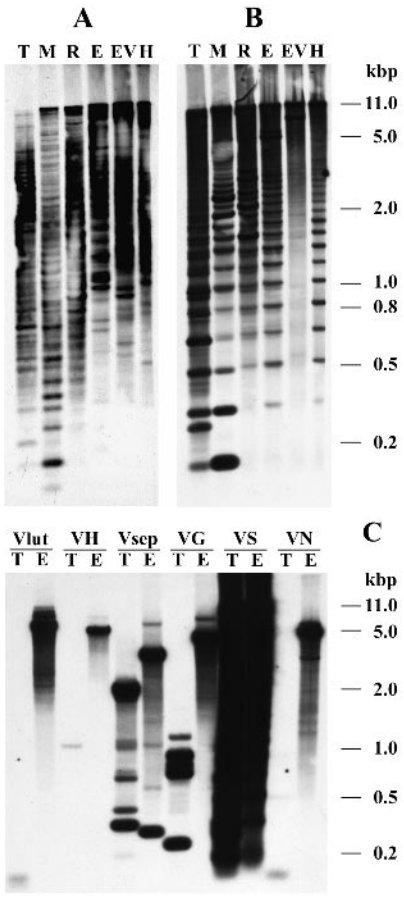


Figure 3. Genomic organization of N6 and S12 repeats. Southern blots of *V. narbonensis* (A) and *V. sativa* (B) genomic DNAs digested with various restriction enzymes (T, *Taq*I; M, *Mbo*I; R, *Rsa*I; E, *Eco*RI; EV, *Eco*RV; H, *Hin*dIII) were probed with the repeat N6 (A) and S12 (B), respectively. Exposure time 10 min. C. Southern blot of genomic DNAs digested either with *Taq*I (T) or *Eco*RI (E) hybridized with S12 probe. Exposure time 1 h. For abbreviations of species names, see Table 1.

Distribution of individual repeats among Vicia *species*

The abundance of the newly described repeats was usually similar in species from the same taxonomic sections (compare Tables 1 and 4) and was accompanied by similarities for the corresponding Southern blot hybridization patterns in these species. This was particularly obvious in sections Peregrinae and Hypechusa (Figure 2A, B). However, in the latter case, most of the clones isolated from *V. melanops* hybridized differently with *V. lutea* and the remaining three species of the section Hypechusa (*V. melanops, V. hybrida, V. pannonica*), giving a much lower signal intensity in *V. lutea* (Table 4 and Figure 2C).

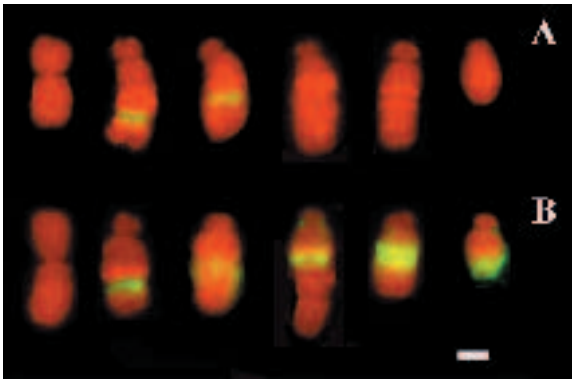


Figure 4. Localization of rDNA and S12 sequences on metaphase chromosomes of *Vicia sativa*. *In situ* hybridization was performed using probes for rDNA genes (Ver17) (A) and for newly isolated tandem repeat S12 (B). The probes were detected using streptavidin-fluorescein (yellow-green color); chromosomes were counterstained with DAPI (displayed in red color for better contrast with the fluorescein signal). One chromosome of each type ($n = 6$) is shown. Bar $= 1$ μm.

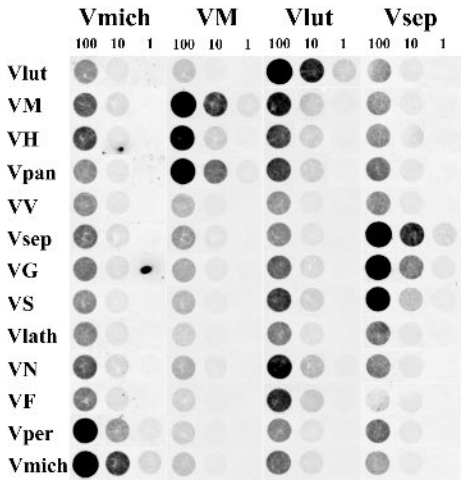


Figure 5. Mutual cross-hybridization of genomic DNAs among *Vicia* species. Dot blots of genomic DNAs of 13 *Vicia* species were hybridized with labeled DNAs of *V. michauxii* (Vmich), *V. melanops* (VM), *V. lutea* (Vlut), and *V. sepium* (Vsep). The order of the genomic DNAs on the dot blots is given on the left, and the amounts of immobilized DNA (100, 10, and 1 ng) are indicated above. Genomic DNAs used for hybridizations are specified at the top of the figure. For abbreviations of the other species names, see Table 1.

To test whether these findings obtained with individual clones represent a more general phenomenon that includes majority of the genomic sequences, dot-blot hybridization experiments with genomic DNAs were performed (Figure 5). As expected, two species from the Peregrinae section (*V. peregrina* and *V. michauxii*) showed very high cross-hybridization, as well as *V. melanops*, *V. hybrida*, and *V. pannonica* (Figure 5, columns Vmich and VM, respectively). The results also proved that *V. lutea* does not show preferential cross-hybridization to the Hypechusa species, as visible from the blots probed with *V. melanops* and *V. lutea* DNA (Figure 5, columns VM and Vlut). Another interesting finding was the close similarity of *V. sepium* to the species from the Vicia section (Figure 5, column Vsep).

Discussion

DNA microarray technologies have been developed mainly as tools for monitoring gene expression in man, yeast, and higher plants (Schena *et al.*, 1995, 1996; Shalon *et al.*, 1996). In this work, we attempted to modify the technique for comparative analysis of repetitive genomic DNA sequences among selected *Vicia* species. Comparison of the hybridization profiles obtained using DNA microarrays and conventional dot-blots revealed agreement in 80% of clones, thus confirming suitability of the microarray approach for preliminary screening. Discrepancies were observed only in clones giving low hybridization signals, and probably resulted from inaccuracies introduced by array hybridization or signal detection and quantification. Recent improvements in DNA microarray-based techniques and in the associated hardware and software (Castellino, 1997; Gwynne and Page, 1999) may eliminate these discrepancies.

Classification of clones into four groups according to their patterns of hybridization with the DNA of different species allowed selection of representative clones for detailed analysis. Sequences representing highly conserved repetitive elements were expected in Group I, which contained clones hybridizing with all tested legume species. Thus, genes coding for ribosomal RNA were found amongst these clones, together with parts of the chloroplast genome (isolates of total genomic DNA inevitably contain some chloroplast sequence contamination). However, no other repeats known to have a structural or functional role in the genome (such as centromeric repeats, or matrix-attachment regions; (Vogt, 1992) were found, suggesting their relatively low abundance and/or sequence variability between the species.

Retroelements are ubiquitous in plant genomes (Flavell *et al.*, 1992; Voytas *et al.*, 1992; Suoniemi *et al.*, 1998), and this fact was reconfirmed by the observation that all Group II clones (hybridizing to most of the *Vicia* and *Pisum* species) comprised

retroelement-like sequences. Although most of these clones contained parts of the relatively conserved reverse transcriptase or integrase coding sequences, they displayed only low cross-hybridization with species outside the *Vicia* and *Pisum* genera. This implies sequence divergence of retroelements present within evolutionarily less related species. An interesting observation was that all retroelement-like clones showed higher homologies to Ty3*/gypsy* than to Ty1*/copia* retrotransposons, suggesting that the former elements are more abundant in the species used in this study.

Group III and IV clones, present in high copy numbers in a limited number or only in a single *Vicia* species, were mostly dispersed repeats of unknown origin. From these only four clones isolated from *V. melanops* (M26, M32, M36, M39) gave positive results in a GenBank search; however, they all displayed homology to an uncharacterized sequence located upstream of pea pectin methylesterase (rcpme1) gene (GenBank AF081457). Since Southern blots probed with M32, M36 and M39 showed several bands common for all three clones (data not shown), it is likely that they are parts of a larger repeat present in pea as well as in the *V. melanops* genome. Several clones (M6, S11, N14, N38, N9, N37, N3, N21) contained two or three tandemly organized subrepeats (33 to 72 bp long). Subrepeats of this type have been described as parts of retroelements (Marschalek *et al.*, 1992; Grandbastien *et al.*, 1994), and suggest that these clones probably contain less conserved retroelement sequences, such as LTRs (SanMiguel *et al.*, 1998), regions between the integrase and the 3′ LTR of Ty3/*gypsy* elements (Chavanne *et al.*, 1998), or leader sequences present between 5′ LTR and *gag* (Suoniemi *et al.*, 1997; Marillonnet and Wessler, 1998).

The occurrence of tandem repeats amongst the Group III and IV clones was expected, since these sequences are often species- or genus-specific (Schmidt and Heslop-Harrison, 1993; King *et al.*, 1995; Maggini *et al.*, 1995; Nakajima *et al.*, 1996; Alix *et al.*, 1998; Nouzová *et al.*, 1999; Staginnus *et al.*, 1999; Macas *et al.*, 2000). Besides clone S38, which was homologous to the previously described VicTR-B repeat family, one new tandemly repeated sequence was isolated from the *V. sativa* genome (S12). Southern blot and *in situ* hybridization experiments indicated that most of the S12 sequences are located outside rDNA genes, forming distinct bands on several chromosomes. Similar satellite sequences showing homology to rDNA spacer subrepeats have been described also in *Vicia faba*, *Vigna radiata* and *Phaseolus vulgaris* (Maggini *et al.*, 1991; Unfried *et al.*, 1991; Falquet *et al.*, 1997). However, although they have similar monomer sizes, they share only partial (61% in *V. faba*) or no detectable homology with the S12 sequence.

The genus *Vicia* is a very complex taxon in which it is difficult to assess inter-species relationships more precisely by conventional taxonomic, cytological, and hybridization studies (Raina and Ogihara, 1995). Our data indicate that some of the newly isolated repetitive sequences could be used as supportive molecular markers for phylogenetic reconstruction of this genus in a manner similar to that used for the genera *Triticum*, *Avena*, *Oryza*, *Hordeum*, and *Pisum* (Dvořák and Zhang, 1992; Solano *et al.*, 1992; Zhao and Kochert, 1993; Svitashev *et al.*, 1994; Ellis *et al.*, 1998). Distribution of most sequences followed the current taxonomic classification of the species within both the Fabaceae family and the genus *Vicia*. Specific cross-hybridization of many of the *Vicia* retroelement-like sequences with *Pisum* species (Table 2) was in agreement with the assignment of both genera to the tribe Viciae (Polhill and Raven, 1981). The hybridization data also agreed with the assignment of *Vicia* species to individual sections within the genus, with the exception of *V. lutea* that considerably differed from the other three species from the section Hypechusa. The hybridization data also suggested a close relationship between *V. sepium* (section Attosa) and two species from the section Vicia, namely *V. grandiflora* and *V. sativa*.. This is in agreement with data obtained by Potokina *et al.* (1999) based on RAPD and chloroplast DNA polymorphisms.

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